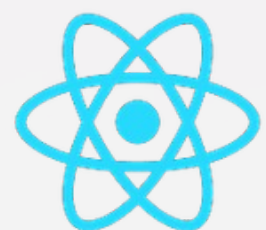


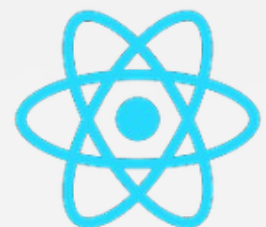
AI Mobile Engineer

What if your app could remember API data, instantly update the UI before the server responds, and automatically refresh stale data in the background?



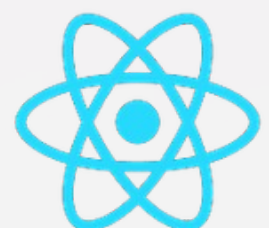
What is TanStack Query?

- A library for managing server state
- Handles fetching, caching, synchronization and updates
- Reduces manual API state management
- Works perfectly with REST APIs



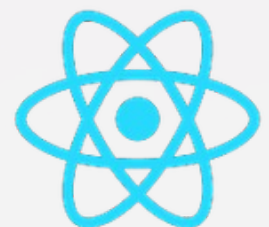
Server State

- Data that comes from your backend/server
- Examples: users, products, posts, orders
- Can become outdated
- Needs fetching, caching and synchronization



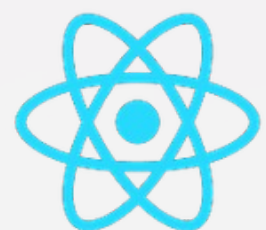
Caching

- Stores previously fetched server data
- Avoids unnecessary API requests
- Makes screens load faster
- Cached data can be refreshed when needed



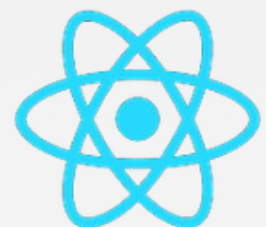
Optimistic Updates

- Update UI before server confirmation
- Makes apps feel instant
- If request fails, rollback the change
- Common for likes, toggles, edits and deletes



Background Refetch

- Refreshes stale data automatically
- Happens without blocking the user
- Keeps app data up to date
- Useful when returning to a screen/app



Why TanStack Query?

- Loading & error states
- Caching
- Refetching
- Mutations
- Retry handling
- Query synchronization

